

Lecture 4

TCP/IP (DoD) Model

Subject: Computer Networking

Introduction to TCP/IP Model

The TCP/IP model is a networking model that explains how data is communicated from one device to another over a network. It describes the different tasks performed when data travels from the sender to the receiver.

TCP/IP stands for Transmission Control Protocol/Internet Protocol.

The TCP/IP model is also commonly called the DoD model.

DoD stands for Department of Defense. The model was developed from networking research associated with the U.S. Department of Defense.

The TCP/IP model consists of four main layers:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

The data moves downward through these layers at the sender side and upward through the layers at the receiver side.

Each layer performs a specific function and adds the information required for its job.

1. Application Layer

Definition:

The Application Layer is the top layer of the TCP/IP model. It provides network services that applications use to communicate with other applications over a network.

The application itself is not the Application Layer. Instead, applications use protocols of the Application Layer for network communication.

Major Functions:

- Provides network services to applications.
- Allows applications to communicate over a network.
- Defines how application data is formatted and exchanged.

Important Protocols:

HTTP – Hypertext Transfer Protocol

HTTP is used for communication between web browsers and web servers. It is commonly used for accessing web pages.

HTTPS – Hypertext Transfer Protocol Secure

HTTPS is the secure version of HTTP. It provides encrypted communication between a browser and a web server.

FTP – File Transfer Protocol

FTP is used to transfer files between computers over a network.

SMTP – Simple Mail Transfer Protocol

SMTP is mainly used for sending email messages between mail systems.

DNS – Domain Name System

DNS converts domain names such as `www.example.com` into IP addresses so that computers can locate the required server.

Simple idea:

The Application Layer answers:

”What network service does the application need?”

2. Transport Layer

Definition:

The Transport Layer provides end-to-end communication between applications or processes running on different devices.

It makes sure that data can be delivered to the correct application at the receiving device and provides different types of delivery services.

The two major Transport Layer protocols are TCP and UDP.

TCP

TCP – Transmission Control Protocol

TCP is a connection-oriented and reliable transport protocol. It provides ordered delivery of data and uses different mechanisms to handle problems during communication.

TCP is commonly used when reliable delivery is important, such as web communication, email, and file transfer.

UDP

UDP – User Datagram Protocol

UDP is a connectionless transport protocol. It has less overhead and does not guarantee delivery or correct ordering of data.

UDP is useful for applications where low delay and simplicity are important, such as online gaming, real-time communication, and some streaming applications.

Port Numbers

The Transport Layer uses port numbers to identify the particular application or process that should receive the data.

For example:

- Port 80 – HTTP
- Port 443 – HTTPS
- Port 53 – DNS
- Port 25 – SMTP

An IP address helps identify the destination device, while a port number helps identify the application or service on that device.

Multiplexing

Multiplexing means collecting data from multiple applications on the sender's device and sending it through the Transport Layer.

Port numbers are used to keep the data of different applications identifiable.

Demultiplexing

Demultiplexing is the reverse process. At the receiving device, the Transport Layer uses the destination port number to deliver the data to the correct application or process.

Error Control

Error control helps deal with problems such as lost or corrupted data.

TCP uses acknowledgements, sequence numbers, and retransmission to provide reliable delivery.

Flow Control

Flow control prevents a fast sender from sending data faster than the receiver can handle.

It helps prevent the receiver's buffer from becoming overloaded.

Congestion Control

Congestion control helps control the amount of traffic entering the network.

TCP uses congestion control mechanisms to reduce network congestion when necessary.

Simple idea:

The Transport Layer answers:

"Which application should receive the data, and what type of delivery should be provided?"

3. Internet Layer

Definition:

The Internet Layer is responsible for logical addressing and routing packets between different networks.

The main protocol of this layer is IP.

IP

IP – Internet Protocol

IP provides logical addresses for network communication. An IP address identifies the logical destination of a packet and allows routers to forward the packet toward its destination.

For example:

Source IP: 192.168.1.10

Destination IP: 8.8.8.8

The Internet Layer mainly performs:

- Logical addressing
- Packet delivery between networks
- Routing

Routing

Routing is the process of determining where a packet should be forwarded so that it can move toward its destination.

Routers examine the destination IP address and use routing information to decide the next direction for the packet.

Simple idea:

The Internet Layer answers:

”Where does this packet need to go?”

4. Network Access Layer

Definition:

The Network Access Layer is the lowest layer of the TCP/IP model. It is responsible for communication over the local network and the actual transmission of frames through network technologies such as Ethernet and Wi-Fi.

This layer is also commonly called the Link Layer.

MAC Address

MAC – Media Access Control

A MAC address is a link-layer address associated with a network interface.

It is used for communication between devices on the same local network or link.

For example:

A4:5E:60:12:AB:91

MAC addresses are important for local delivery.

For example, when a computer sends data to a router, the frame uses the MAC address of the router on that local link.

When the router forwards the packet to the next network, a new frame is created for the next link.

Therefore, MAC addresses are mainly concerned with the current local link, while IP addresses are used for logical addressing and routing across networks.

Ethernet and Wi-Fi

Ethernet and Wi-Fi are common technologies used at the Network Access Layer.

Ethernet is commonly used for wired network communication.

Wi-Fi is commonly used for wireless network communication.

Simple idea:

The Network Access Layer answers:

”How will the data be delivered over the current local network?”

Encapsulation

When data is sent, it moves downward through the layers.

For example, suppose an application wants to send the message:

Hello

At the Application Layer, the message is created as data.

At the Transport Layer, TCP or UDP information is added. When TCP is used, the resulting unit is commonly called a segment.

At the Internet Layer, an IP header containing source and destination IP information is added. The result is commonly called a packet.

At the Network Access Layer, link-layer information such as MAC addresses is added. The result is commonly called a frame.

The general process is:

```
Data
|
v
Transport Header + Data
|
v
IP Header + Transport Data
|
v
Link Header + IP Packet
|
v
Frame
```

This process of adding information as data moves down the layers is called **encapsulation**.

At the receiving device, the layers process the information in the reverse direction. This process is called **decapsulation**.

Simple Difference Between Port, IP and MAC

These three terms are very important for understanding networking.

Port Number:

Identifies the application or process that should receive the data.

IP Address:

Provides logical addressing and helps identify the destination for communication across networks.

MAC Address:

Provides link-layer addressing and is used for delivery over the current local network or link.

A simple way to remember them is:

Port = Which application?

IP = Which logical destination?

MAC = Which device on the current local link?

Summary of the Four Layers

Application Layer

Provides network services to applications.

Examples: HTTP, HTTPS, FTP, SMTP, DNS.

Transport Layer

Provides end-to-end communication between applications.

Main protocols: TCP and UDP.

Important concepts: port numbers, multiplexing, demultiplexing, error control, flow control, and congestion control.

Internet Layer

Provides logical addressing and routing between networks.

Main protocol: IP.

Important concepts: IP address and routing.

Network Access Layer

Provides communication over the local network or link.

Important concepts: MAC addresses, Ethernet, and Wi-Fi.

Final Concept

The complete communication process can be remembered as:

Application

|

v

Transport

|

v

Internet

|

v

Network Access

|

v

Physical Network

In simple words:

The Application Layer provides the network service, the Transport Layer provides end-to-end communication between applications, the Internet Layer provides logical addressing and routing, and the Network Access Layer provides delivery over the local network.